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Cosine Similarity Tutorial

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Cosine Similarity Tutorial

Abstract – This is a tutorial on the cosine similarity measure. Its meaning in the context of uncorrelated and orthogonal variables is examined.

Keywords: cosine similarity, tutorial, dot products, vectors, orthogonal, uncorrelated

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Introduction

In previous tutorials we discussed the difference between distance and similarity measures and the risks of arbitrarily transforming or averaging these (Garcia, 2015a; 2015b; 2015c; 2015d).

We mentioned that a *Pearson's Correlation Coefficient* (r) computed from mean-centered variables, or from z-scores, is a cosine similarity. In this case, r can be computed from the regression curve slope.

We also mentioned that like correlations, slopes, and cosines, cosine similarity measures are not additive so we cannot compute arithmetic averages from any of these measures. The same can be said about standard deviations, rates, and dissimilar ratios. It should be noted that a mean value is not an estimate of central tendency when a distribution is either skewed or Cauchy. Furthermore, the Law of Large Numbers does not apply to a Cauchy Distribution.

Conventions used in this tutorial

This tutorial was written as a companion for our Cosine Similarity Calculator (Garcia, 2015a). It may serve as a basic tutorial for students and those starting in data mining and information retrieval. For the sake of clarity, we adopt the following conventions:

- $\mathbf{a}$ and $\mathbf{b}$ are the $\vec{a}$ and $\vec{b}$ vectors.
- dp_{ab} is the $\vec{a} \cdot \vec{b}$ dot product between $\mathbf{a}$ and $\mathbf{b}$.
- dp_{aa} is the dot product of $\mathbf{a}$ with itself.
- dp_{bb} is the dot product of $\mathbf{b}$ with itself.
- l_a and l_b are the $\|\vec{a}\|$ and $\|\vec{b}\|$ vector lengths.
- $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ are unit vectors; i.e., $l_a = l_b = 1$.

Instead of just saying that the cosine similarity between two vectors is given by the expression

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \|\vec{b}\|} \quad (1)$$

we want to explain **what is actually scored** with (1).

Similarity is an interesting measure as there are many ways of computing it. Indeed, we built a tool that computes over 70 different similarity measures (Garcia, 2016). Since there are so many ways of expressing similarity, what kind of resemblance a cosine similarity actually scores? This is the question that this tutorial pretends to address.

Generally speaking, similarity is a measure of resemblance; i.e., how similar or alike things being compared are. One way of computing similarity is through the use of vectors.

Representing data sets as vectors

A *matrix* is just a table filled with values. Suppose that said table consists of r number of rows and c number of columns. We may refer to these as *vectors*. Thus for a square matrix, one with same number of rows and columns, row vectors are data sets of size $n = c$ and column vectors are data sets of size $n = r$.

A vector is a quantity or phenomenon with two independent properties: direction and magnitude. For $n = 2$ and $n = 3$, we can visualize a vector in its n -dimensional space as a line segment ending in an arrow. The orientation of the line segment is its direction, and the length is its magnitude. For $n > 3$ we cannot visualize vectors, but we can still compute them.

Computing Dot Products

We may multiply a vector by *itself or another* vector and compute a quantity called the dot product (*dp*). This is done by multiplying vector elements and taking summations. To illustrate, suppose that $\mathbf{a}$ and $\mathbf{b}$ are vectors such that

- $\mathbf{a} = [1, 2, 3]$
- $\mathbf{b} = [4, -5, 6]$

In this case, the dot product of

- $\mathbf{a}$ with $\mathbf{b}$ is $dp_{ab} = 1*4 + 2*-5 + 3*6 = 12$
- $\mathbf{a}$ with itself is $dp_{aa} = 1*1 + 2*2 + 3*3 = 14$
- $\mathbf{b}$ with itself is $dp_{bb} = 4*4 + -5*-5 + 6*6 = 77$

We may now ask: What kind of information is stored in dp_{ab} , dp_{aa} , and dp_{bb} ?

What information is stored in dp_{ab} ?

Good question: dp_{ab} holds information about the *direction* of the vectors. To be precise, if

- $dp_{ab} > 0$, $\mathbf{a}$ and $\mathbf{b}$ form an angle less than 90° .
- $dp_{ab} = 0$, $\mathbf{a}$ and $\mathbf{b}$ form an angle that is exactly 90° .
- $dp_{ab} < 0$, $\mathbf{a}$ and $\mathbf{b}$ form an angle greater than 90° .

Table 1 shows different types of angles.

Table 1. Possible types of angles.

Type	acute	right	obtuse	straight	reflex	perigon
Angle	$\theta < 90^\circ$	$\theta = 90^\circ$	$180 > \theta > 90^\circ$	$\theta = 180^\circ$	$360^\circ > \theta > 180^\circ$	$\theta = 360^\circ$

An angle of 0° means that $\cos \theta = 1$ so the vectors point to identical directions. An angle of 90° means that $\cos \theta = 0$ so the vectors point to perpendicular directions or are *orthogonal*.

What information is stored in dp_{aa} and dp_{bb} ?

Taking square roots, it is clear that dp_{aa} and dp_{bb} hold *length information*. So

- $l_a = (dp_{aa})^{1/2} = (14)^{1/2} = 3.74$; i.e., the length of $\mathbf{a}$.
- $l_b = (dp_{bb})^{1/2} = (77)^{1/2} = 8.77$; i.e., the length of $\mathbf{b}$.
- $l_a * l_b = (dp_{aa})^{1/2} * (dp_{bb})^{1/2} = 32.83$; i.e., the length product (lp_{ab}) of $\mathbf{a}$ and $\mathbf{b}$.

In other words, (1) is a dot product/length product ratio

$$\cos \theta = \frac{dp_{ab}}{lp_{ab}} = \frac{dp_{ab}}{\sqrt{dp_{aa}}\sqrt{dp_{bb}}} = \frac{12}{32.84} = 0.37 \quad (2)$$

Therefore, when we compute a cosine similarity we are measuring the direction-length resemblance between data sets represented as vectors.

Comparing Vectors of Different Lengths

To compare vectors of different lengths, these can be recomputed as *unit vectors*. A unit vector is computed by dividing its elements by its length. In other words, we are re the previous vectors as

$$\hat{a} = [1/3.74, 2/3.74, 3/3.74]$$

$$\hat{b} = [4/8.77, -5/8.77, 6/8.77]$$

where the hat (^) denotes a unit vector. Since the new lengths are equal to 1, the cosine similarity between $\hat{a}$ and $\hat{b}$ is their dot product; hence

$$\cos \theta = dp_{\hat{a}\hat{b}} = 0.37 \quad (3)$$

Expressions (1), (2), and (3) return the same result, confirming that a cosine is a judgment of the orientation of the vectors, independent of their lengths (Wikipedia, 2015a).

What does $\cos \theta = 0$ mean?

As mentioned before, an angle of 90° means that $\cos \theta = 0$ so the vectors are perpendicular or orthogonal. This does not necessarily mean that the variables are uncorrelated.

According to Rodgers, Nicewander, & Toothaker (1984) when referring to variables, orthogonal denotes that *raw* variables are perpendicular while uncorrelated means that *centered* variables are perpendicular. Centered variables are those with their mean removed, so they have zero mean.

It turns out that unlike vector length normalization, subtracting the mean from raw variables can change the angle between the vectors. The following can happen to the raw variables:

- If they are perpendicular, can become not perpendicular so they are orthogonal, but not uncorrelated.
- If they are not perpendicular, can become perpendicular so they are uncorrelated, but not orthogonal.
- If they are perpendicular and remain perpendicular, they are orthogonal and uncorrelated.
- If they are not perpendicular and remain not perpendicular, they are neither orthogonal nor uncorrelated—although their angle can change.

The following figure, adapted from Rodgers *et. al* (1984), illustrates these relationships. Notice that not all uncorrelated variables are orthogonal and *vice versa* (Wikipedia, 2015b). The figure also shows that while all uncorrelated variables are independent, the reverse is not true.

Therefore, if a textbook states that orthogonal variables are uncorrelated, it probably refers to paired random variables with zero mean; i.e., to the overlapping region shown in the figure.

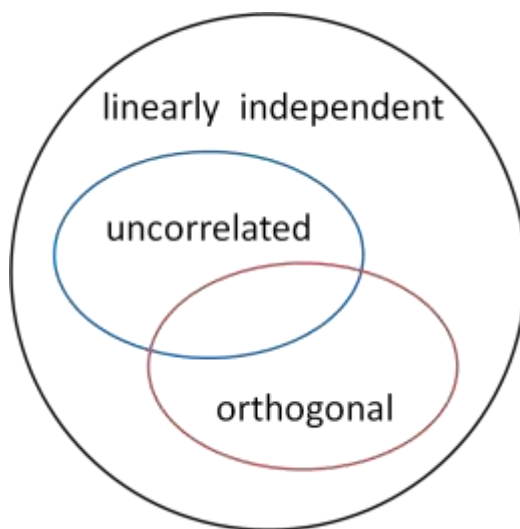


Figure 1. Venn Diagram for linearly independent, uncorrelated, and orthogonal variables.

These results are not surprising. It can be shown for centered variables that

$$r = \cos \theta \quad (4)$$

Because r is the covariance of paired variables normalized by their standard deviation,

$$r = \frac{COV_{ab}}{S_a S_b} \quad (5)$$

it must follow for *centered* variables that if $\cos \theta = 0$, then $r = 0$ and $COV_{ab} = 0$.

The relationship $r = \cos \theta$ is also true for standardized variables, also known as z-scores. These are centered variables normalized by their standard deviation. Transforming variables into z-scores is, though not always, useful and recommended.

Once in a z-score format, we can run other types of tests on the variables like a quantile-quantile analysis (Garcia, 2015e), or even compute r , and therefore $\cos \theta$, from the slope of the regression curve of z-scores. For those interested, we have developed another tool called the Standardizer (Garcia, 2015f). This tool was initially thought of as an *x-to-z-score* standardizer, hence its name. It is now a versatile statistical tool for univariate analysis.

Conclusion

Unlike other similarity measures, a cosine similarity is a measure of the direction-length resemblance between vectors.

An angle of 0° means that $\cos \theta = 1$ and that the vectors are oriented in identical directions; i.e., that the corresponding data sets are completely similar to one another. An angle of 90° means that $\cos \theta = 0$ and that the corresponding variables are perpendicular, but not necessarily that are uncorrelated.

Computing $\cos \theta$ from raw and center variables are two different things. This fact can be used to examine relationships between paired variables. Regardless of the method used to compute cosine similarities, and for the sake of transparency, it is always a good idea to state whether raw or centered variables were used and why.

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